

lab 7

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```
library(ggplot2)
library(tidyverse)
```

```
## — Attaching core tidyverse packages — tidyverse 2.0.0 —
## ✓ dplyr      1.1.4      ✓ readr      2.1.5
## ✓ forcats    1.0.0      ✓ stringr    1.5.1
## ✓ lubridate  1.9.3      ✓ tibble     3.2.1
## ✓ purrr      1.0.2      ✓ tidyr      1.3.1
## — Conflicts — tidyverse_conflicts() —
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts
to become errors
```

```
library(haven)
```

```
setwd("~/Documents/Masters/Econometrics/data")# your directory structure will be different
load("ACS_2021_couples.RData")

#summary(acs2021_couples)
```

```
acs2021_couples$RACE <- fct_recode(as.factor(acs2021_couples$RACE),
                                   "White" = "1",
                                   "Black" = "2",
                                   "American Indian or Alaska Native" = "3",
                                   "Chinese" = "4",
                                   "Japanese" = "5",
                                   "Other Asian or Pacific Islander" = "6",
                                   "Other race" = "7",
                                   "two races" = "8",
                                   "three races" = "9")

acs2021_couples$h_race <- fct_recode(as.factor(acs2021_couples$h_race),
                                     "White" = "1",
                                     "Black" = "2",
                                     "American Indian or Alaska Native" = "3",
                                     "Chinese" = "4",
                                     "Japanese" = "5",
                                     "Other Asian or Pacific Islander" = "6",
                                     "Other race" = "7",
                                     "two races" = "8",
                                     "three races" = "9")

acs2021_couples$HISPAN <- fct_recode(as.factor(acs2021_couples$HISPAN),
                                      "Not Hispanic" = "0",
                                      "Mexican" = "1",
                                      "Puerto Rican" = "2",
                                      "Cuban" = "3",
                                      "Other" = "4")

acs2021_couples$h_hispan <- fct_recode(as.factor(acs2021_couples$h_hispan),
                                       "Not Hispanic" = "0",
                                       "Mexican" = "1",
                                       "Puerto Rican" = "2",
                                       "Cuban" = "3",
                                       "Other" = "4")
```

```
trad_data <- acs2021_couples %>% filter( (SEX == "Female") & (h_sex == "Male") )
trad_data$he_more_than_10yrs_than_her <- as.numeric(trad_data$sage_diff <= -10)
```

```
ols_out1 <- lm(he_more_than_10yrs_than_her ~ educ_hs + educ_somecoll + educ_college + ed
uc_advdeg + AGE, data = trad_data)
summary(ols_out1)
```

```
##
## Call:
## lm(formula = he_more_than_10yrs_than_her ~ educ_hs + educ_somecoll +
##      educ_college + educ_advdeg + AGE, data = trad_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.18687 -0.09641 -0.07386 -0.05087  0.99776
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   2.129e-01  2.138e-03   99.57  <2e-16 ***
## educ_hs       -3.348e-02  1.767e-03  -18.95  <2e-16 ***
## educ_somecoll -4.461e-02  1.854e-03  -24.06  <2e-16 ***
## educ_college  -6.600e-02  1.835e-03  -35.98  <2e-16 ***
## educ_advdeg   -7.068e-02  1.934e-03  -36.55  <2e-16 ***
## AGE           -1.628e-03  2.606e-05  -62.47  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.269 on 412269 degrees of freedom
## Multiple R-squared:  0.01302,    Adjusted R-squared:  0.013
## F-statistic: 1087 on 5 and 412269 DF,  p-value: < 2.2e-16
```

```
data_kansas_missouri <- trad_data %>%
  filter(STATEFIP %in% c("Kansas", "Missouri"))

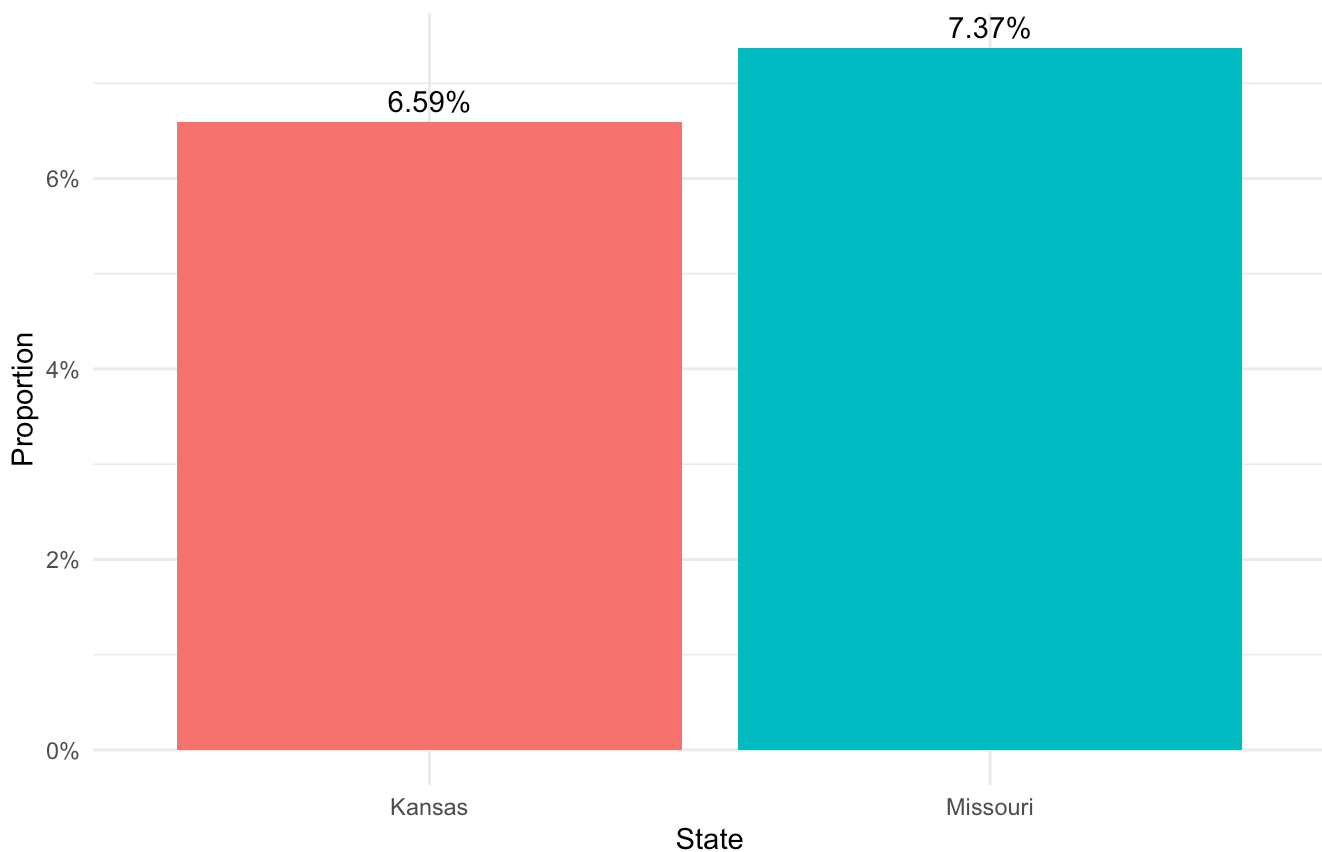
data_kansas_missouri$STATEFIP <- fct_drop(data_kansas_missouri$STATEFIP)
summary(data_kansas_missouri$STATEFIP)
```

```
##    Kansas Missouri
##      3778      8016
```

```
data_kansas_missouri_age_gap_mean <- data_kansas_missouri %>%
  group_by(STATEFIP) %>%
  summarize(
    prop_10plus_gap = mean(he_more_than_10yrs_than_her),
    n = n()
  )

ggplot(data_kansas_missouri_age_gap_mean, aes(x = STATEFIP, y = prop_10plus_gap, fill =
STATEFIP)) +
  geom_col() +
  geom_text(aes(label = paste0(round(prop_10plus_gap * 100, 2), "%")),
    vjust = -0.5) +
  labs(title = "Proportion of Couples Where Man is 10+ Years Older",
    subtitle = "Kansas vs Missouri",
    x = "State",
    y = "Proportion") +
  scale_y_continuous(labels = scales::percent) +
  theme_minimal() +
  theme(legend.position = "none")
```

Proportion of Couples Where Man is 10+ Years Older
Kansas vs Missouri



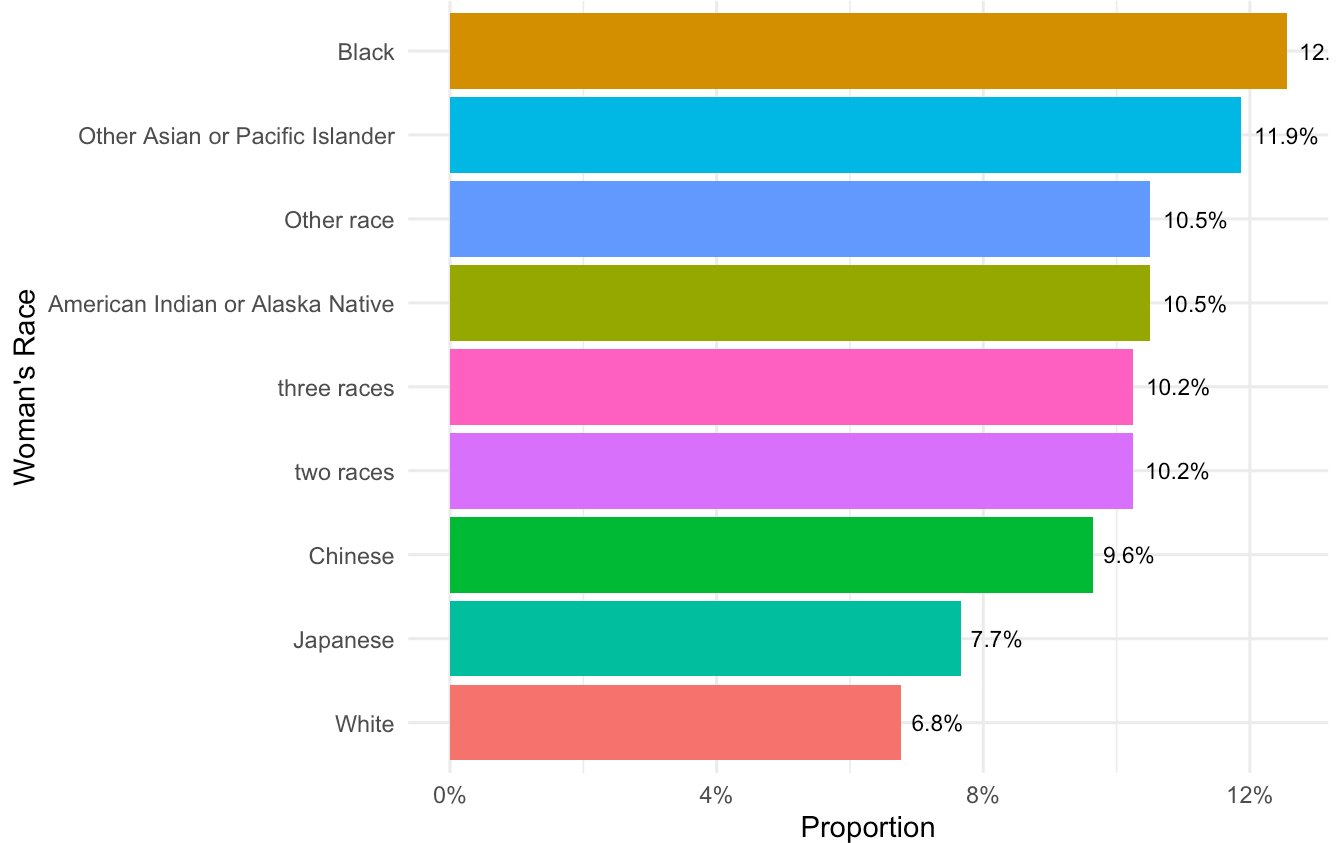
```
table(trad_data$RACE)
```

```
##
##           White           Black
##           294610         19252
## American Indian or Alaska Native           Chinese
##           3772           8482
##           Japanese Other Asian or Pacific Islander
##           2037           26443
##           Other race           two races
##           21298           34537
##           three races
##           1844
```

```
age_gap_by_race <- trad_data %>%
  group_by(RACE) %>%
  summarize(
    prop_10plus_gap = mean(he_more_than_10yrs_than_her),
    n = n()
  ) %>%
  arrange(desc(prop_10plus_gap))

ggplot(age_gap_by_race, aes(x = reorder(RACE, prop_10plus_gap), y = prop_10plus_gap, fill = RACE)) +
  geom_col() +
  geom_text(aes(label = paste0(round(prop_10plus_gap * 100, 1), "%"),
    hjust = -0.2, size = 3) +
  coord_flip() +
  labs(title = "Proportion of Couples Where Man is 10+ Years Older",
    subtitle = "By Woman's Race (US-wide)",
    x = "Woman's Race",
    y = "Proportion") +
  scale_y_continuous(labels = scales::percent) + # No limits now
  theme_minimal() +
  theme(legend.position = "none")
```

Proportion of Couples Where Man is 10+ Years Older By Woman's Race (US-wide)

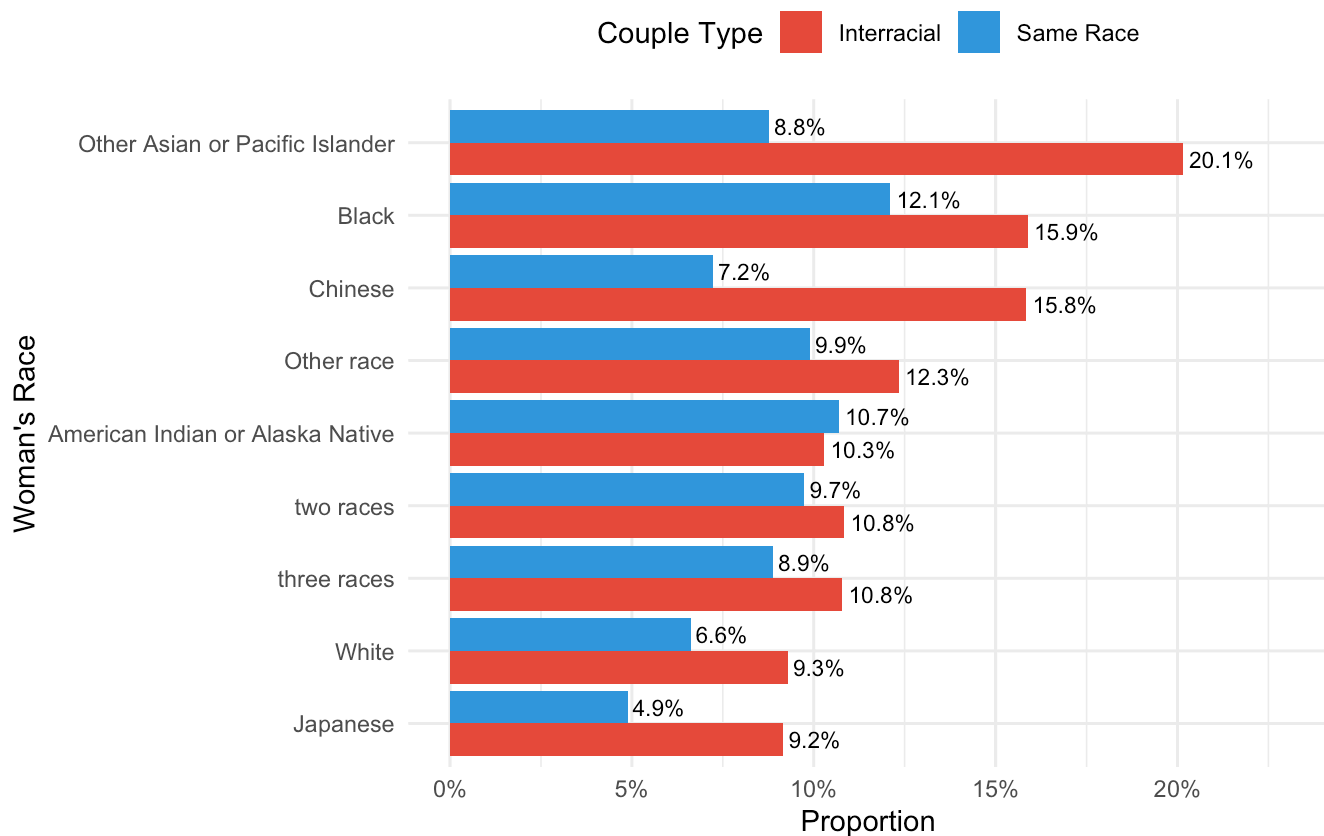


```
trad_data$same_race <- as.numeric(trad_data$RACE == trad_data$h_race)
race_interracial <- trad_data %>%
  mutate(couple_type = ifelse(same_race == 1, "Same Race", "Interracial")) %>%
  group_by(RACE, couple_type) %>%
  summarize(
    prop_10plus_gap = mean(he_more_than_10yrs_than_her),
    n = n()
  ) %>%
  filter(n >= 100)
```

```
## `summarise()` has grouped output by 'RACE'. You can override using the
## `.groups` argument.
```

```
ggplot(race_interracial, aes(x = reorder(RACE, prop_10plus_gap), y = prop_10plus_gap, fill = couple_type)) +
  geom_col(position = "dodge") +
  geom_text(aes(label = paste0(round(prop_10plus_gap * 100, 1), "%")),
            position = position_dodge(width = 0.9),
            hjust = -0.1, size = 3) +
  coord_flip() +
  labs(title = "Proportion of Couples Where Man is 10+ Years Older",
        subtitle = "By Woman's Race and Interracial Status",
        x = "Woman's Race",
        y = "Proportion",
        fill = "Couple Type") +
  scale_y_continuous(labels = scales::percent, limits = c(0, 0.23)) +
  scale_fill_manual(values = c("Interracial" = "#E74C3C", "Same Race" = "#3498DB")) +
  theme_minimal() +
  theme(legend.position = "top")
```

Proportion of Couples Where Man is 10+ Years Older
By Woman's Race and Interracial Status



```
lm <- lm(data = trad_data, formula = he_more_than_10yrs_than_her ~ same_race + RACE + h_race)
summary(lm)
```

```
##
## Call:
## lm(formula = he_more_than_10yrs_than_her ~ same_race + RACE +
##     h_race, data = trad_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.21111 -0.08777 -0.06629 -0.06629  0.96838
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      0.096440   0.001425   67.679 < 2e-16
## same_race      -0.030153   0.001413  -21.345 < 2e-16
## RACEBlack       0.036255   0.003746   9.679 < 2e-16
## RACEAmerican Indian or Alaska Native  0.018476   0.005254   3.516 0.000438
## RACEChinese     0.066765   0.004920  13.570 < 2e-16
## RACEJapanese    0.003036   0.006838   0.444 0.657013
## RACEOther Asian or Pacific Islander  0.092225   0.002962  31.139 < 2e-16
## RACEOther race   0.033057   0.003179  10.398 < 2e-16
## RACEtwo races    0.024795   0.001881  13.184 < 2e-16
## RACEthree races  0.024630   0.006630   3.715 0.000203
## h_raceBlack     0.022448   0.003624   6.194 5.86e-10
## h_raceAmerican Indian or Alaska Native 0.011763   0.005433   2.165 0.030378
## h_raceChinese   -0.058523   0.005350 -10.940 < 2e-16
## h_raceJapanese  -0.019857   0.008852  -2.243 0.024884
## h_raceOther Asian or Pacific Islander -0.064823   0.003192 -20.308 < 2e-16
## h_raceOther race -0.001643   0.003341  -0.492 0.622939
## h_racetwo races -0.003313   0.001887  -1.756 0.079053
## h_racethree races -0.025873   0.006657  -3.886 0.000102
##
## (Intercept)      ***
## same_race        ***
## RACEBlack        ***
## RACEAmerican Indian or Alaska Native ***
## RACEChinese      ***
## RACEJapanese     ***
## RACEOther Asian or Pacific Islander  ***
## RACEOther race   ***
## RACEtwo races    ***
## RACEthree races  ***
## h_raceBlack      ***
## h_raceAmerican Indian or Alaska Native *
## h_raceChinese     ***
## h_raceJapanese    *
## h_raceOther Asian or Pacific Islander ***
## h_raceOther race
## h_racetwo races  .
## h_racethree races ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2696 on 412257 degrees of freedom
```



```
## Multiple R-squared:  0.008429,   Adjusted R-squared:  0.008388  
## F-statistic: 206.1 on 17 and 412257 DF,  p-value: < 2.2e-16
```